Section 3.1 IM-OLS Cointegration Estimation Matrix: U.S. NFC Transformation Elasticity (1929–2024)

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Data Provenance and Estimation Summary

This report presents the empirical cointegration results for Section 3.1 of Chapter 2. All models are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over the full NIPA sample window (1931–2024, $T = 94$ observations after listwise deletion for the lag-1 conditioning variables ω_{t-1} and ι_{t-1}). Point estimates are tuning-parameter free. To ensure exact size control under the non-linear integration orders of the regressors ($I(1) \times I(1)$ and $I(1) \times I(1) \times I(1)$) and account for break-search degrees of freedom loss, significance stars ($^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$) for all coefficient rows ($\hat{\alpha}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual VW Type 2 ($S_{\text{Type 2}}$ / CT_{IM}) cointegration test are computed using a **100,000-pass custom Monte Carlo simulation suite** tailored to each model's exact DGP specification. Distributional conditioning variables (ω_{t-1} and ι_{t-1}) are centered at their 1929 initial values, preserving $\hat{\theta}_0$ as the capital elasticity at the initial state. VIF statistics are computed on the level regressors space. Step level-shift dummies $D_{T_b}^{\text{step}}$ represent structural breaks identified via a first-pass Bai-Perron residual audit (15% trimming) and are dynamically adjusted within the custom critical value DGP.

1 Specification Set 1: Double Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1}$)

Table 1: Specification Set 1 (Part A: Baseline NFC & Corporate GVA Wage Shares): Double Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
Demand Control	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share
$\hat{\alpha}$ (Intercept)	6.47*	8.97	6.47*	8.97	6.29*	6.81	6.29*	6.81	6.48*	9.19	6.48*	9.19
	(0.34)	(0.89)	(0.34)	(0.89)	(0.34)	(0.83)	(0.34)	(0.83)	(0.35)	(0.98)	(0.35)	(0.98)
$\hat{\theta}_0^{\text{priv}}$ (Private Core k_{priv})	0.59**	0.34	0.59**	0.34	—	—	—	—	0.59**	0.33	0.59**	0.33
	(0.02)	(0.08)	(0.02)	(0.08)					(0.03)	(0.09)	(0.03)	(0.09)
$\hat{\theta}_0^{\text{aug}}$ (Infra-Augmented k_{aug})	—	—	—	—	0.59**	0.52	0.59**	0.52	—	—	—	—
					(0.02)	(0.07)	(0.02)	(0.07)				
$\hat{\theta}_1$ (Interaction)	0.02	-0.05	0.02	-0.05	0.06	-0.07	0.06	-0.07	0.01	-0.07	0.01	-0.07
	(0.13)	(0.15)	(0.13)	(0.15)	(0.12)	(0.10)	(0.12)	(0.10)	(0.12)	(0.16)	(0.12)	(0.16)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	1.38	1.38	1.38	1.38	1.40	1.40	1.40	1.40	1.54	1.54	1.54	1.54
VIF (Interaction)	1.38	1.38	1.38	1.38	1.40	1.40	1.40	1.40	1.54	1.54	1.54	1.54
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.15**	-3.00	-4.15**	-3.00	-3.97**	-3.52*	-3.97**	-3.52*	-4.10**	-3.15	-4.10**	-3.15
Shin LM (C_{Shin}) [H_0: Coint.]	2.12***	0.56***	2.12***	0.56***	2.15***	0.92***	2.15***	0.92***	2.08***	0.45***	2.08***	0.45***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	819.23***	1835.00***	819.23***	1835.00***	924.93***	719.49***	924.93***	719.49***	934.98***	1794.32***	934.98***	1794.32***
Bandwidth Ratio (b)	0.2852	0.1737	0.2852	0.1737	0.2832	0.1946	0.2832	0.1946	0.2876	0.1658	0.2876	0.1658
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows ($\hat{\alpha}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual VW Type 2 ($S_{\text{Type 2}}$ / CT_{IM}) cointegration test are derived from a **100,000-pass custom Monte Carlo simulation suite** mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, and with/without deterministic $I(0)$ demand controls). This custom simulation adjusts the critical values for the non-linear integration orders of the regressors ($I(1) \times I(1)$ and $I(1) \times I(1) \times I(1)$) and accounts for break-search degrees of freedom loss under the dynamic Bai-Perron audit (15% trimming). Resid. ADF stars follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8) to whiten serial correlation. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184) and are included for reference. **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0).

Table 2: Specification Set 1 (Part B: Shaikh-Tonak Tier 2 Wage Share): Double Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M1.7a	M1.7b	M1.8a	M1.8b	M1.9a	M1.9b	M1.10a	M1.10b	M1.11a	M1.11b	M1.12a	M1.12b
Demand Control	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share
$\hat{\alpha}$ (Intercept)	6.29*	8.75	6.29*	8.75	7.64	5.44	7.64	5.44	7.52	4.84	7.52	4.84
	(0.35)	(0.90)	(0.35)	(0.90)	(0.67)	(1.23)	(0.67)	(1.23)	(0.69)	(1.24)	(0.69)	(1.24)
$\hat{\theta}_0^{\text{priv}}$ (Private Core k_{priv})	—	—	—	—	0.48	0.67	0.48	0.67	—	—	—	—
					(0.06)	(0.11)	(0.06)	(0.11)				
$\hat{\theta}_0^{\text{aug}}$ (Infra-Augmented k_{aug})	0.59**	0.35	0.59**	0.35	—	—	—	—	0.48	0.70	0.48	0.70
	(0.02)	(0.08)	(0.02)	(0.08)					(0.06)	(0.10)	(0.06)	(0.10)
$\hat{\theta}_1$ (Interaction)	0.05	-0.04	0.05	-0.04	-0.28	-0.19	-0.28	-0.19	-0.27	-0.18	-0.27	-0.18
	(0.12)	(0.14)	(0.12)	(0.14)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	1.56	1.56	1.56	1.56	8.04	8.04	8.04	8.04	8.14	8.14	8.14	8.14
VIF (Interaction)	1.56	1.56	1.56	1.56	8.04	8.04	8.04	8.04	8.14	8.14	8.14	8.14
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-3.92**	-2.95	-3.92**	-2.95	-3.40	-2.78	-3.40	-2.78	-3.13	-2.86	-3.13	-2.86
Shin LM (C_{Shin}) [H_0: Coint.]	2.11***	0.60***	2.11***	0.60***	0.76***	0.20***	0.76***	0.20***	0.91***	0.20***	0.91***	0.20***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	1004.15***	2306.76***	1004.15***	2306.76***	743.93***	2169.68***	743.93***	2169.68***	713.58***	1404.04***	713.58***	1404.04***
Bandwidth Ratio (b)	0.2856	0.1704	0.2856	0.1704	0.2665	0.2105	0.2665	0.2105	0.2830	0.1907	0.2830	0.1907
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows ($\hat{\alpha}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual VW Type 2 ($S_{\text{Type 2}}$ / CT_{IM}) cointegration test are derived from a **100,000-pass custom Monte Carlo simulation suite** mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, and with/without deterministic $I(0)$ demand controls). This custom simulation adjusts the critical values for the non-linear integration orders of the regressors ($I(1)\times I(1)$ and $I(1)\times I(1)\times I(1)$) and accounts for break-search degrees of freedom loss under the dynamic Bai-Perron audit (15% trimming). Resid. ADF stars follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8) to whiten serial correlation. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184) and are included for reference. **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0).

2 Specification Set 2: Triple Interactions ($k^{\mathbf{cap}} \cdot \omega_{t-1} \cdot \iota_{t-1}$)

Table 3: Specification Set 2 (Part A: Baseline NFC & Corporate GVA Wage Shares): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
Demand Control	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share
$\hat{\alpha}$ (Intercept)	6.46	8.18	9.39	9.41	6.20	8.08	9.14	7.22	6.50	7.76	9.32	9.36
	(0.49)	(1.06)	(0.77)	(0.88)	(0.50)	(1.10)	(0.79)	(0.88)	(0.50)	(1.16)	(0.75)	(0.89)
$\hat{\theta}_0^{\text{priv}}$ (Private Core k_{priv})	0.59	0.42	0.51	0.36	—	—	—	—	0.59	0.45	0.51	0.36
	(0.04)	(0.09)	(0.03)	(0.07)					(0.04)	(0.10)	(0.03)	(0.07)
$\hat{\theta}_0^{\text{aug}}$ (Infra-Augmented k_{aug})	—	—	—	—	0.60	0.41	0.52	0.50	—	—	—	—
					(0.04)	(0.09)	(0.03)	(0.06)				
$\hat{\theta}_1$ (Interaction)	0.00	-0.19	-0.06	-0.09	0.04	-0.17	-0.03	-0.46	-0.01	-0.16	-0.06	-0.07
	(0.10)	(0.09)	(0.07)	(0.08)	(0.10)	(0.10)	(0.07)	(0.13)	(0.10)	(0.10)	(0.07)	(0.08)
$\hat{\beta}_{\text{cm}}$ (NetC^{share})	—	—	-3.45	-0.93	—	—	-3.42	-0.25	—	—	-3.37	-0.92
			(0.76)	(0.62)			(0.77)	(0.60)			(0.75)	(0.63)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	3.57	3.57	4.41	4.41	3.69	3.69	4.61	4.61	3.86	3.86	4.64	4.64
VIF (Interaction)	3.57	3.57	3.58	3.58	3.69	3.69	3.71	3.71	3.86	3.86	3.87	3.87
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-4.02**	-4.00**	-2.11	-2.64	-3.82**	-3.00	-2.16	-3.21	-3.97**	-3.93**	-2.10	-2.62
Shin LM (C_{Shin}) [H_0: Coint.]	1.86***	0.66***	2.08***	0.83***	1.90***	0.71***	2.08***	0.41***	1.78***	0.72***	2.08***	0.83***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3910.69***	276.16***	—	—	3724.48***	359.15***	—	—	4021.16***	140.50***	—	—
Bandwidth Ratio (b)	0.2908	0.1688	NA	NA	0.2916	0.1688	NA	NA	0.2934	0.1734	NA	NA
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows ($\hat{\alpha}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual VW Type 2 ($S_{\text{Type 2}}$ / CT_{IM}) cointegration test are derived from a **100,000-pass custom Monte Carlo simulation suite** mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, and with/without deterministic $I(0)$ demand controls). This custom simulation adjusts the critical values for the non-linear integration orders of the regressors ($I(1) \times I(1)$ and $I(1) \times I(1) \times I(1)$) and accounts for break-search degrees of freedom loss under the dynamic Bai-Perron audit (15% trimming). Resid. ADF stars follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8) to whiten serial correlation. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184) and are included for reference. **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0).

Table 4: Specification Set 2 (Part B: Shaikh-Tonak Tier 2 Wage Share): Triple Interaction IM-OLS Cointegration Estimates

Dependent Variable: $y_t^{\mathbf{NFC}}$ (Log NFC Real GVA)												
Model	M2.7a	M2.7b	M2.8a	M2.8b	M2.9a	M2.9b	M2.10a	M2.10b	M2.11a	M2.11b	M2.12a	M2.12b
Demand Control	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share	None	None	NetC Share	NetC Share
$\hat{\alpha}$ (Intercept)	6.24	7.93	9.06	8.60	8.30	8.68	9.42	9.49	8.15	8.44	9.38	8.98
	(0.52)	(1.13)	(0.78)	(1.00)	(0.81)	(0.70)	(0.70)	(0.81)	(0.87)	(0.73)	(0.74)	(0.88)
$\hat{\theta}_0^{\text{priv}}$ (Private Core k_{priv})	—	—	—	—	0.42	0.39	0.44	0.43	—	—	—	—
					(0.07)	(0.06)	(0.04)	(0.06)				
$\hat{\theta}_0^{\text{aug}}$ (Infra-Augmented k_{aug})	0.60	0.43	0.52	0.41	—	—	—	—	0.43	0.40	0.44	0.45
	(0.04)	(0.10)	(0.03)	(0.08)					(0.07)	(0.06)	(0.05)	(0.07)
$\hat{\theta}_1$ (Interaction)	0.02	-0.15	-0.03	-0.01	-0.31	-0.38	-0.23	-0.30	-0.30	-0.36	-0.20	-0.29
	(0.10)	(0.10)	(0.07)	(0.09)	(0.13)	(0.11)	(0.08)	(0.07)	(0.13)	(0.11)	(0.08)	(0.07)
$\hat{\beta}_{\text{cm}}$ ($\text{NetC}^{\text{share}}$)	—	—	-3.33	-0.89	—	—	-2.21	-2.12	—	—	-2.48	-1.99
			(0.76)	(0.63)			(0.66)	(0.39)			(0.68)	(0.40)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
Collinearity Diagnostics (VIF)												
VIF (k^{cap})	4.00	4.00	4.84	4.84	14.01	14.01	14.05	14.05	14.46	14.46	14.48	14.48
VIF (Interaction)	4.00	4.00	4.01	4.01	14.01	14.01	15.20	15.20	14.46	14.46	15.56	15.56
Cointegration Test Triad												
Resid. ADF (t_{ADF}) [H_0: No Coint.]	-3.77**	-2.98	-2.16	-2.50	-3.27	-2.90	-2.52	-3.06	-2.30	-2.85	-2.47	-3.09
Shin LM (C_{Shin}) [H_0: Coint.]	1.81***	0.76***	2.10***	0.90***	0.96***	0.51***	1.29***	0.38***	1.25***	0.52***	1.31***	0.39***
VW Type 2 ($S_{\text{Type 2}}$) [H_0: Coint.]	3822.34***	223.87***	—	—	88.61**	3570.21***	—	—	91.10**	3285.32***	—	—
Bandwidth Ratio (b)	0.2940	0.1691	NA	NA	0.2713	0.2047	NA	NA	0.2924	0.2029	NA	NA
Sample Size (T)	94	94	94	94	94	94	94	94	94	94	94	94

Note: Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang et al. 2024) over 1929–2024. Standard errors in parentheses. Significance: $^{*}p < 0.10$,$^{**}p < 0.05$,$^{***}p < 0.01$. Stars for coefficient rows ($\hat{\alpha}$, $\hat{\theta}_0$, $\hat{\theta}_1$, $\hat{\beta}_{\text{cm}}$) and the Residual VW Type 2 ($S_{\text{Type 2}}$ / CT_{IM}) cointegration test are derived from a **100,000-pass custom Monte Carlo simulation suite** mapping the exact DGP variant (double vs. triple interaction polynomials, with/without step level-shifts, and with/without deterministic $I(0)$ demand controls). This custom simulation adjusts the critical values for the non-linear integration orders of the regressors ($I(1) \times I(1)$ and $I(1) \times I(1) \times I(1)$) and accounts for break-search degrees of freedom loss under the dynamic Bai-Perron audit (15% trimming). Resid. ADF stars follow MacKinnon (2010) Engle-Granger critical values, indexed by the total number of $I(1)$ variables; the test is lag-augmented with the order selected by AIC (max 8) to whiten serial correlation. Shin LM stars use Shin's (1994) critical values (10%/5%/1% = 0.095/0.121/0.184) and are included for reference. **Star reversal warning:** For Residual ADF, stars indicate rejection of No Cointegration (supporting cointegration). For Shin LM and VW Type 2, stars indicate rejection of Cointegration. Distributional variables centered at 1929 initial state (t_0).